

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for realworld applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I **G. Ram Charan (192525043)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____

Assessment Tasks

Case Study

Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations. **Smart Retail Inventory Management System**

1. Cloud services used:

Cloud storage, Web APIs, platform analytics services, and demand forecasting services are used.

2. Role of Web APIs:

APIs connect POS systems and dashboards with centralized cloud services for data exchange.

3. Role of cloud storage:

It securely stores product data, transaction logs, and analytics datasets.

4. Role of platform services:

Platform services analyze inventory data and support demand forecasting.

5. Security mechanisms:

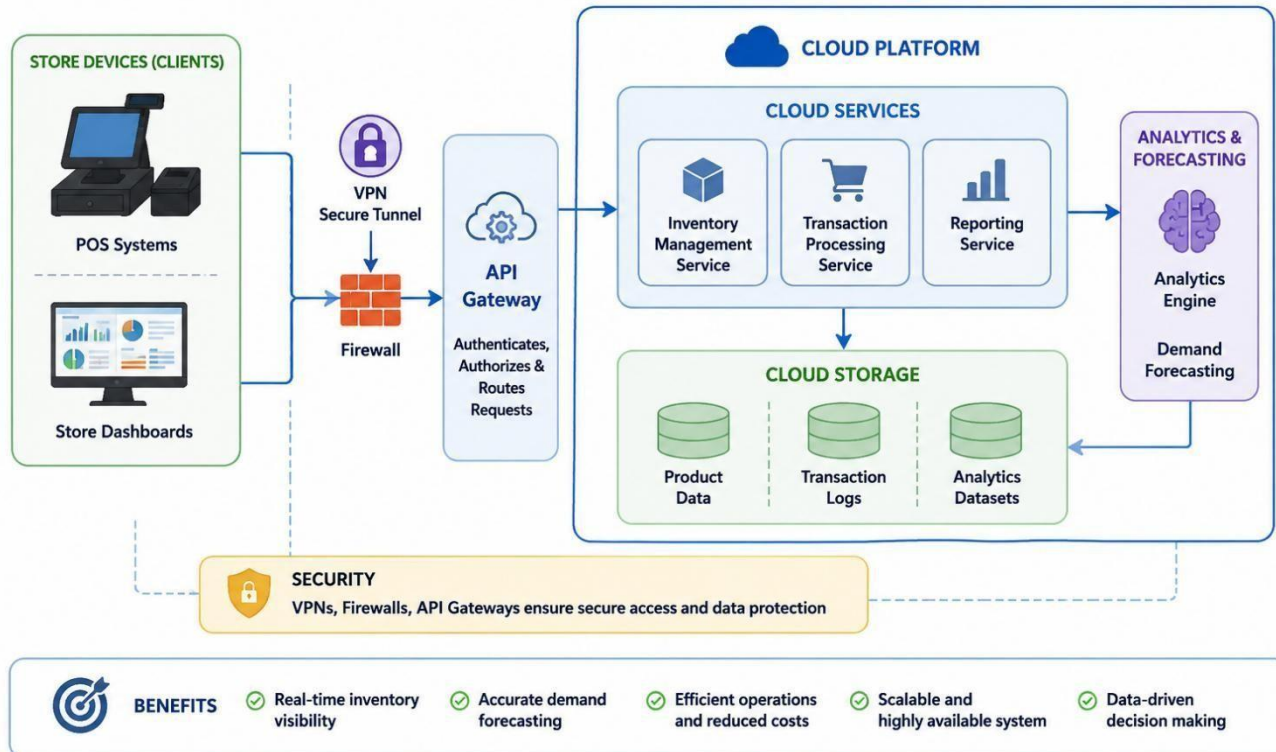
VPNs, firewalls, and API gateways provide secure access to cloud resources.

6. Overall benefit:

The cloud solution improves inventory management, analytics, forecasting, and operational efficiency.

Case Study 1: Smart Retail Inventory Management System

Cloud Architecture



Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration. **Cloud-Based Document Collaboration System**

1. Cloud services used:

Distributed cloud storage, Web APIs, network infrastructure, and security services are used.

2. Role of cloud storage:

Documents are stored in distributed cloud storage with version control.

3. Role of Web APIs:

APIs synchronize documents and changes across multiple clients and devices.

4. Role of browsers:

Users access the platform through browsers for real-time document editing and sharing.

5. Role of network infrastructure:

It provides low latency and high availability for users worldwide.

6. Security mechanisms:

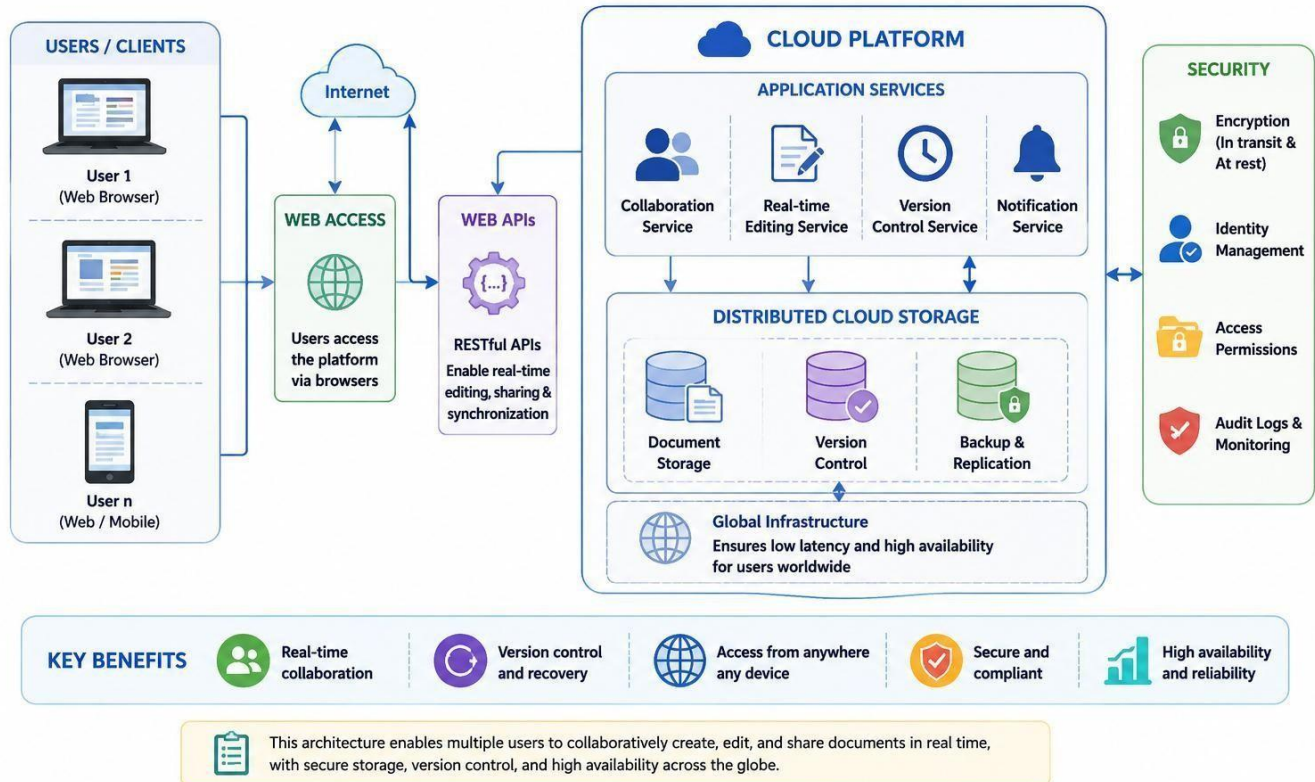
Encryption, identity management, and access permissions protect collaborative documents.

7. Overall benefit:

The system enables secure, real-time, and reliable document collaboration across devices.

Case Study 2: Cloud-Based Document Collaboration System

Architecture Diagram



Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time. **Online Food Ordering Platform**

1. Cloud services used:

Cloud platform services, RESTful Web APIs, cloud storage, virtual machines, and managed Kubernetes are used.

2. Role of RESTful APIs:

REST APIs connect web/mobile clients with backend cloud services.

3. Role of cloud storage:

It securely stores menu images, invoices, and customer data.

4. Role of virtual machines:

Virtual machines provide the computing infrastructure required to run backend services.

5. Role of Kubernetes:

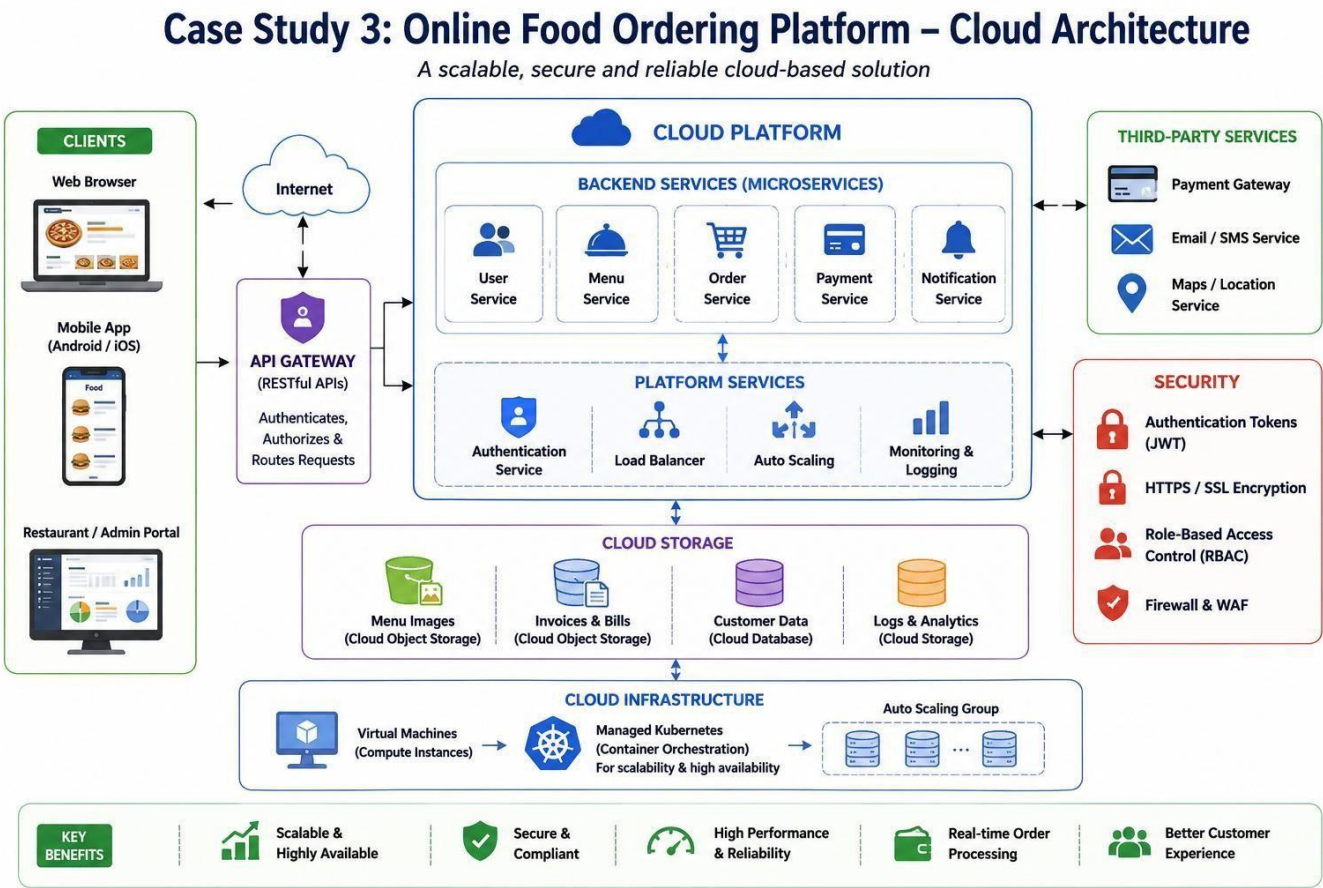
Managed Kubernetes provides scalability and reliability for the application.

6. Security mechanisms:

Authentication tokens, HTTPS, and role-based access control secure the platform.

7. Overall benefit:

The cloud architecture provides a scalable, reliable, secure, and real-time food ordering system.



Case Study	Main Cloud Focus	Key Technologies
1. Smart Retail	Inventory & analytics	APIs, cloud storage, analytics, forecasting
2. Document Collaboration	Real-time collaboration	Distributed storage, APIs, version control
3. Food Ordering	Scalable online service	REST APIs, storage, VMs, Kubernetes

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
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Understanding of Case	25%	Clearly identifies all technical and	Identifies most requirements	Basic understanding only	Poor understanding of case
		business requirements			
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Wellstructured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized